I need some statistics for my paper



Let's just plug my data on Graphpad



Keep scrolling through the tests until



P < 0.001 (***)



Are statistics done terribly wrong ?

Winter school 2023

~~Are statistics done terribly wrong ?~~ Yes

Statistics done wrong

Winter school 2023

Outline

- How badly do we do stats ?
- Start with the basics
- *p* values – and why you probably should just avoid them
- Sample size and a small trick
- Some prescriptions

Disclaimers

- Much of this talk is based on the *excellent* « Point of Significance » series from Nature by Martin Krzywinski & Naomi Altman, and from the book « Statistics Done Wrong: The Woefully Complete Guide » by Alex Reinhart

- My hope is that this presentation is useless to you, and that you already know everything.
- No offense intended, I am guilty of *every* fault I point here.
- I am not a statistician, don’t @ me, don’t ask questions.

Ambitions of this talk – Empowerment promise

- Review the basics of the basics…
- …so that you don’t make the cover of *Science* with handcuffs for data fraud.

- What you won’t get:
  - Explanations of how to select the best statistical test for your experiment
  - Advanced concepts
  - A probability course

How badly we do stats

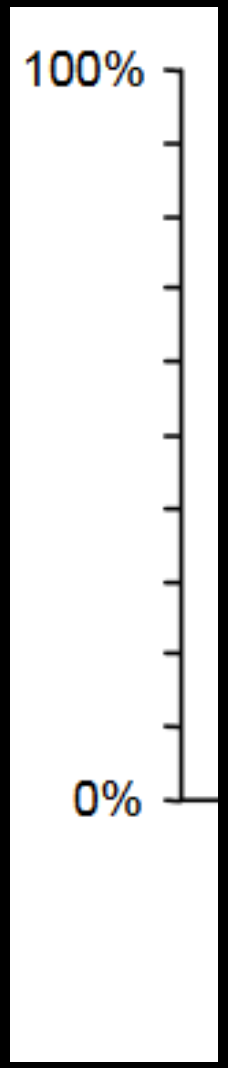
Haller & Krauss – Misinterpretations of Significance: A problem students share with their teachers?

- In 6 German universities – quizzed members of psychology departments with a short questionnaire about statistics.

- Three groups
  - Methodology instructors (N=30) : professors and tutors that teach statistics to psychology students
  - Scientific psychologists (N=39) : professors and other scientific staff not involved in the teaching of statistics
  - Psychology students (N=44)

Methods of Psychological Research Online **7**, 2002

Haha psychologists (~~or haha Germans~~)



Percentage of participants in each group who made at least one mistake

Br. J. Surg. 1988, Vol. 75, July, 664–667

The task of a statistical referee

G. D. Murray

Department of Surgery, Royal Infirmary, 10 Alexandra Parade, Glasgow G31 2ER, UK

Correspondence to: Dr G. D. Murray

The results are presented of an audit of the statistical standard of papers published in The British Journal of Surgery. *A number of deficiencies are highlighted, many of which stem from an over-emphasis on statistical significance at the expense of any assessment of the clinical relevance of research findings. The flaws in the design of published studies, and in particular the many instances of inadequate sample sizes, emphasize that statistical input should be sought at the beginning of a research project rather than at the end.*

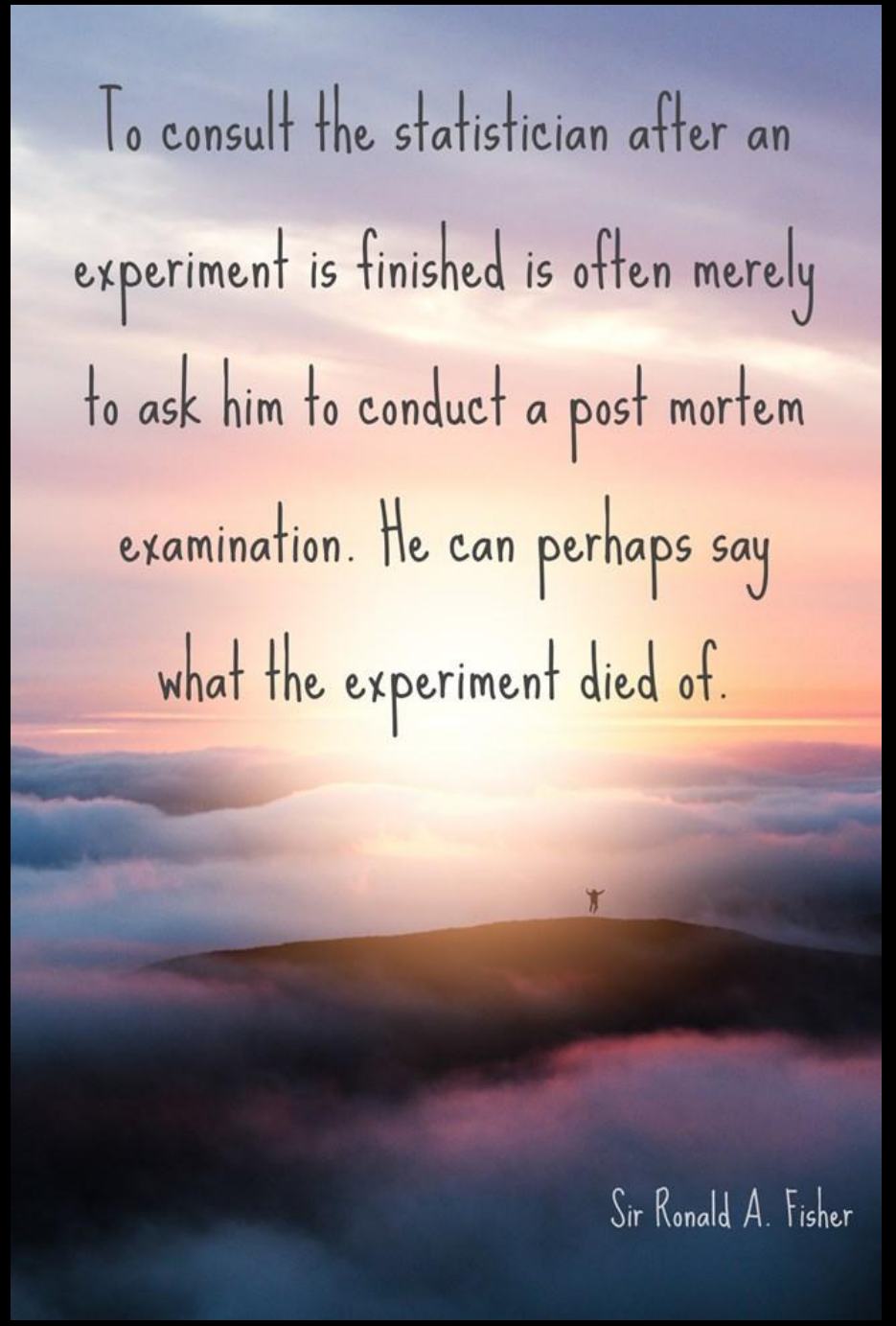
Keywords: Surgical studies, statistics, audit

The task of a statistical referee – G.D Murray, 1988

In recent years there has been increasing recognition of the need to improve the standard of statistical methodology used in medical research. A number of reviews of leading medical journals[1-4] have shown a consistent pattern, with some 50 per cent of published papers containing identifiable statistical errors, of which around 50 per cent are classified as 'serious'.

Problems of design

It is at the design stage that a statistician has the greatest potential to contribute to the ultimate success of a research project. In my own experience, and in that of many of my statistical colleagues, it happens all too rarely that statistical advice is sought at this stage. Rather it is more common to be approached by an investigator who has completed a large data collection exercise, and who is seeking a liberal sprinkling of *P* values to render the study suitable for publication. Indeed the first contact is often made after the paper is returned by a journal with various statistical criticisms, and the statistician is expected to perform a salvage job on a study which more likely than not suffers from a fundamental design problem which not even the most sophisticated of analyses can overcome.



Haha medical doctors

September 5, 2007

Medicine Residents' Understanding of the Biostatistics and Results in the Medical Literature

Donna M. Windish, MD, MPH; Stephen J. Huot, MD, PhD; Michael L. Green, MD, MSc

» Author Affiliations | Article Information

JAMA. 2007;298(9):1010-1022. doi:10.1001/jama.298.9.1010

- Survey both residents and faculty on common statistical concepts
- Residents score on average less than 50% correct
- Faculty score on a average less than 75% correct
- 95% of the residents find it important to understand those concepts

Results The survey was completed by 277 of 367 residents (75.5%) in 11 residency programs. The overall mean percentage correct on statistical knowledge and interpretation of results was 41.4% (95% confidence interval [CI], 39.7%-43.3%) vs 71.5% (95% CI, 57.5%-85.5%) for fellows and general medicine faculty with research training ($P < .001$). Higher scores in

Uh oh



PLOS MEDICINE

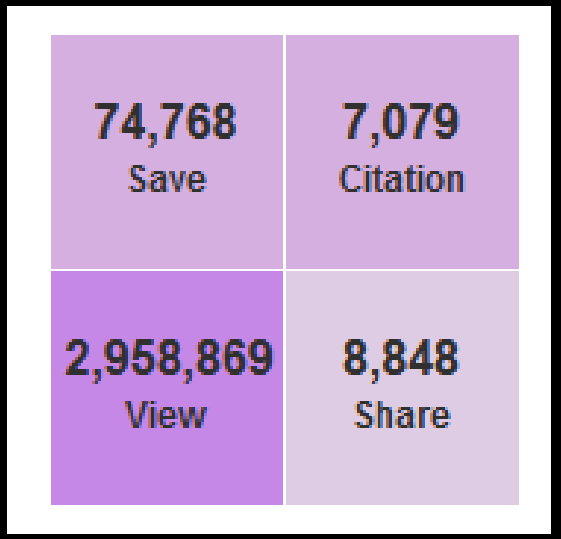
OPEN ACCESS

ESSAY

Why Most Published Research Findings Are False

John P. A. Ioannidis

Published: August 30, 2005 • https://doi.org/10.1371/journal.pmed.0020124



74,768 Save	7,079 Citation
2,958,869 View	8,848 Share

How about you ?

- Suppose you're testing two drugs, Fixitol and Solvix.
- You have two treatment groups, one that takes Fixitol and one that takes Solvix, and you measure their performance on a standard fitness test afterward.
- You compare the mean score of each group using a simple significance test, and you obtain $p = 0.01$, indicating there is a statistically significant difference between means.

https://tweedback.de/z4zj



Statistics: why ?

Start with the basics

Do I really need statistics ?



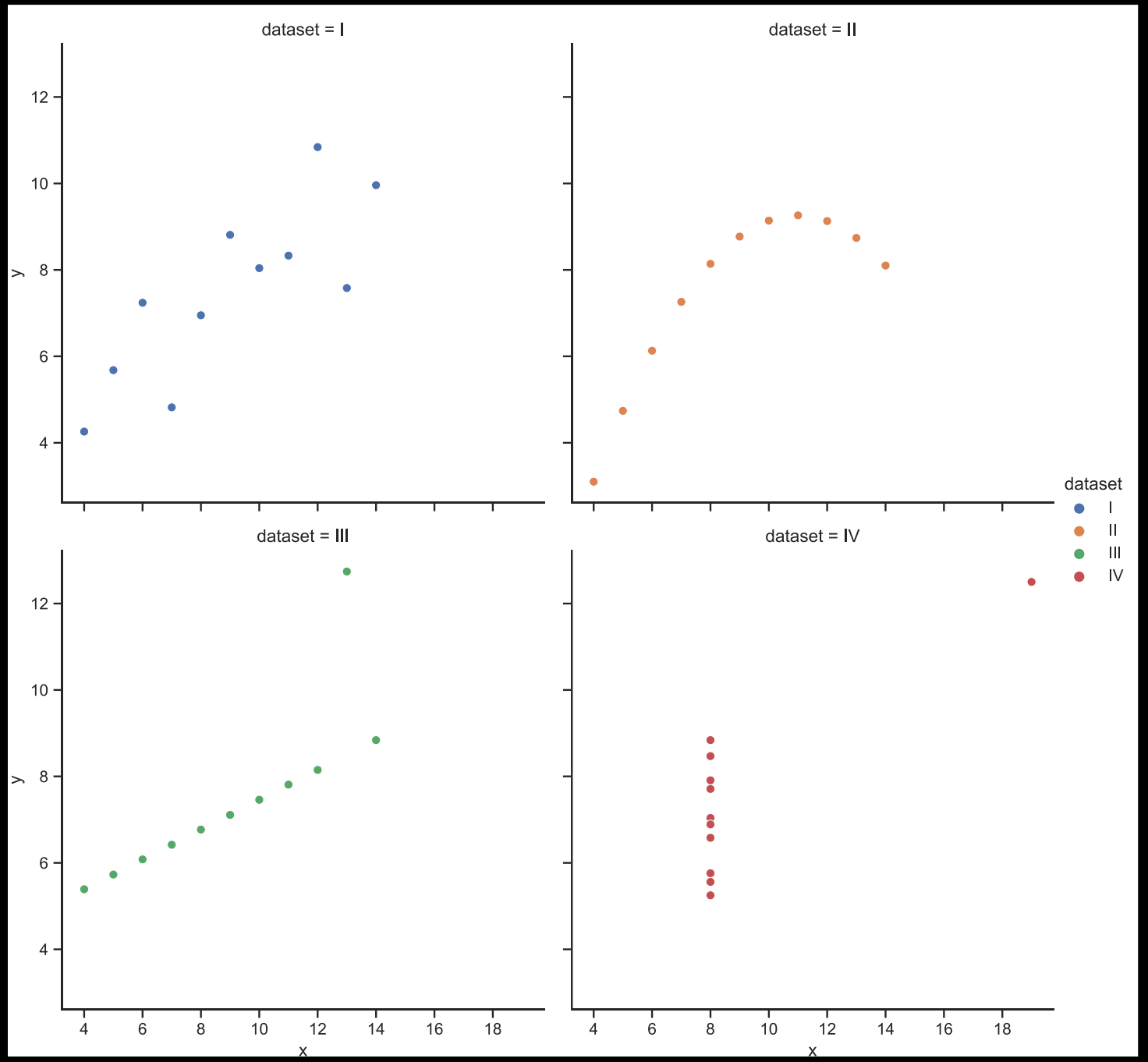
<u>Option 1 – No, but this is not a suggestion :</u>
« S*muel, we need some graphs with stars, otherwise those st*pid biol*gists are going to murder us. »

<u>Option 2 – Descriptive stats :</u>
Summarise the main features of a data set, with measures such as the mean and the standard deviation.

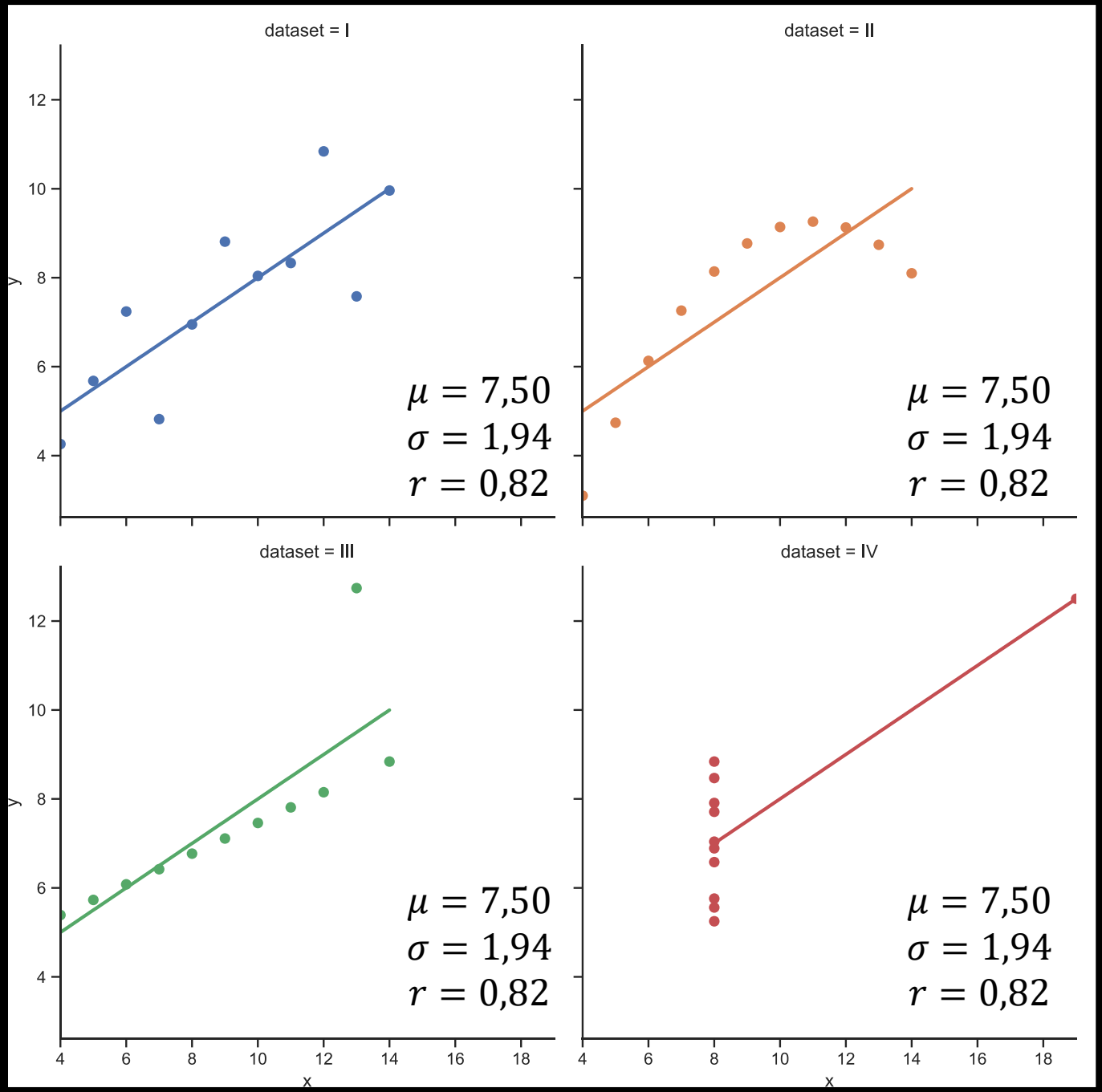
<u>Option 3 – Inferential stats :</u>
Generalise from the *observed data* to the world at large

Faces, names, and professions have been anonymised to preserve reputations.

Data points first, ALWAYS



Data points first, ALWAYS



The Anscombe Quartet

Typical start

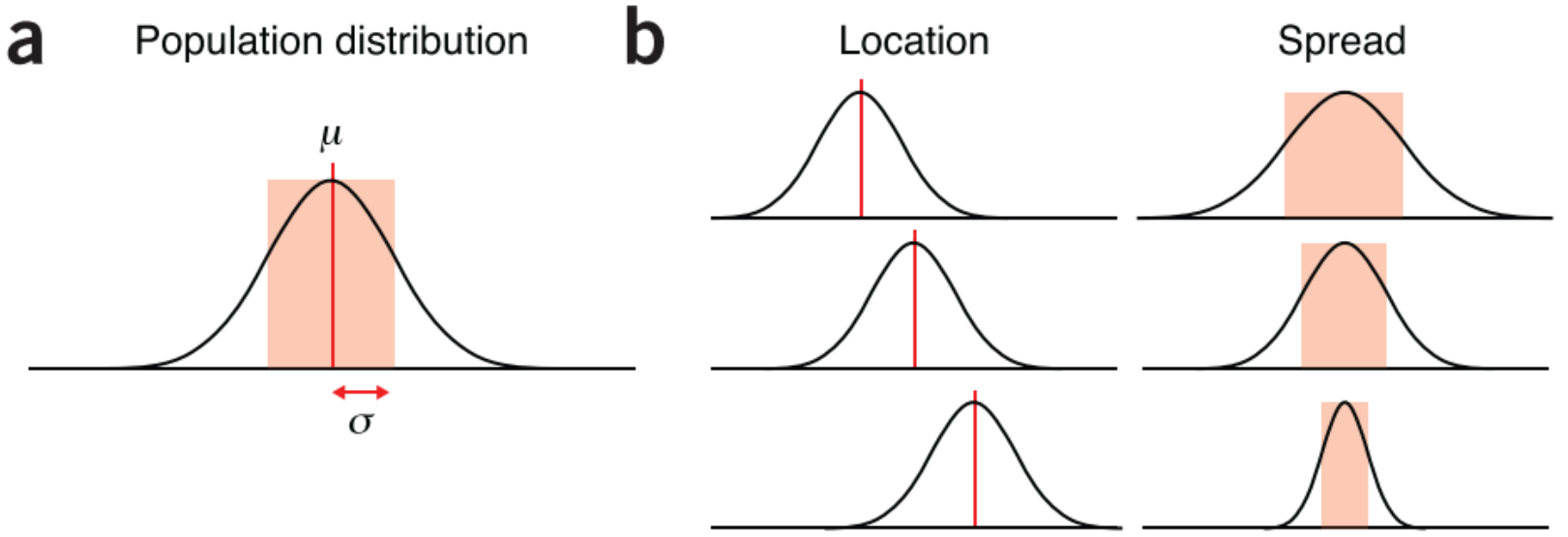
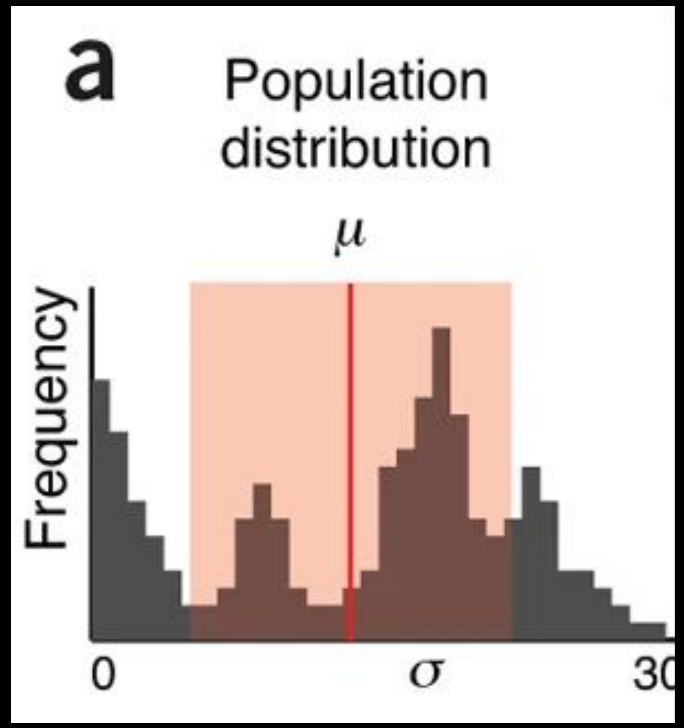


Figure 1 | The mean and s.d. are commonly used to characterize the location and spread of a distribution. When referring to a population, these measures are denoted by the symbols μ and σ.

We generally do not have access to the whole population, so we sample

Note how « *highly informative* » μ and σ are here.



$\mu_{\bar{X}}$ the mean of the sample means, $\sigma_{\bar{X}}$ the s.d. of sample means.

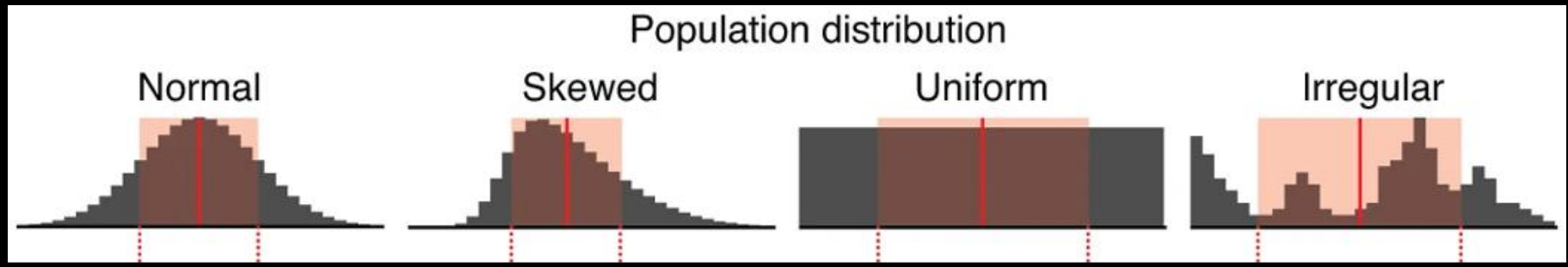
Obligatory slide : the Central Limit Theorem (CLT)

- « The distribution of sample means becomes increasingly close to a normal distribution as the sample size increases, regardless of the shape of the population distribution*. »

* : provided the frequency of extreme values decreases quickly

Obligatory slide : the Central Limit Theorem (CLT)

Draw 10 000 samples of size *n* from the population.



Progressively increase the *size of the samples*.

Assuming unbiased sampling.

The beauty of CLT

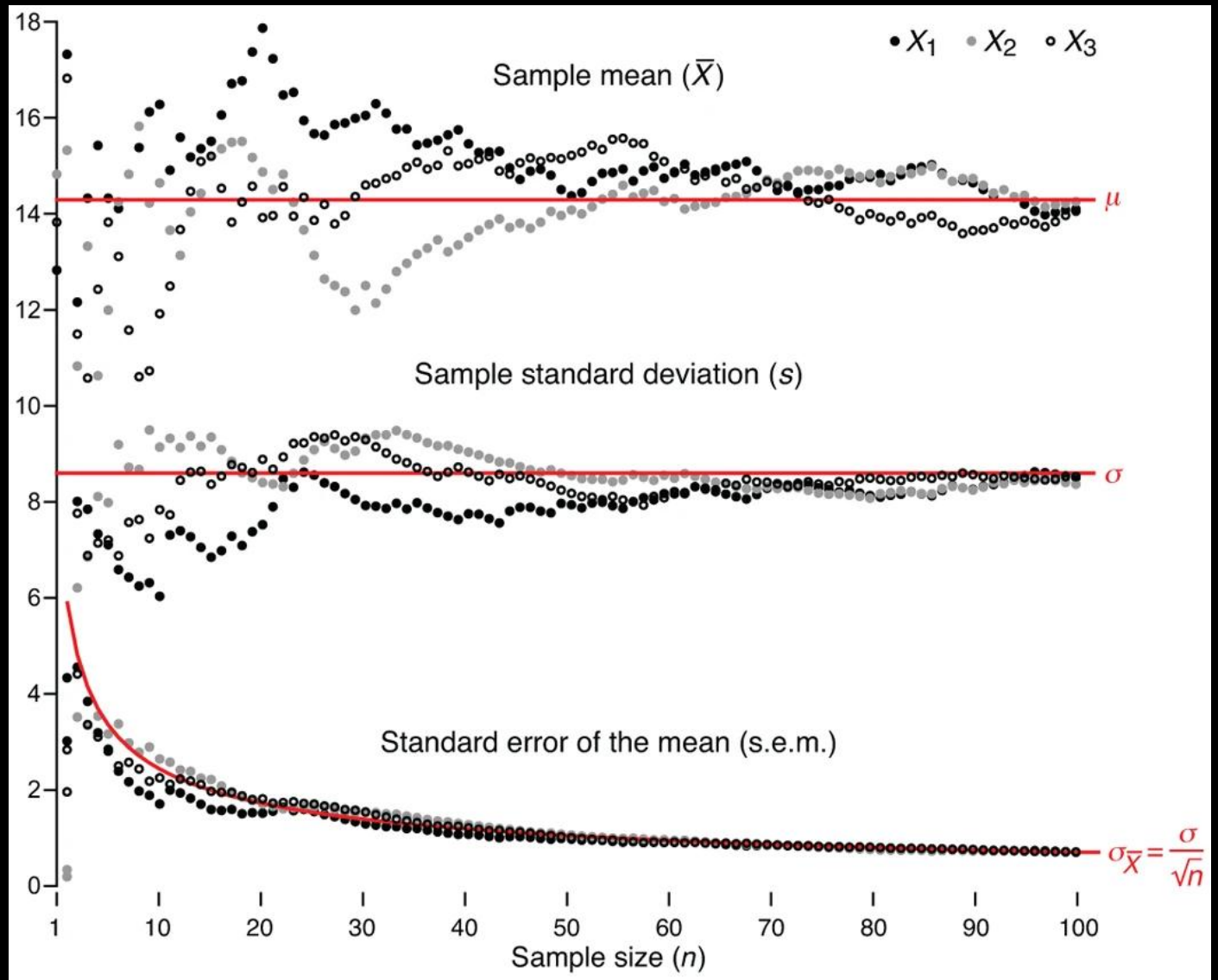
The larger your sample size, the better your sample mean $\bar{X}$ will approximate your population mean μ.

The larger your sample size, the better your sample s.d. s will approximate your population s.d. σ.

The larger your sample size, the more the SPREAD OF YOUR SAMPLE MEAN (the s.e.m.) will decrease.

Careful 1: $\sigma_{\bar{X}} \neq s$ -> the spread of the population stays the same.

Careful 2: to *double* your precision, you need to acquire *four* times more data.

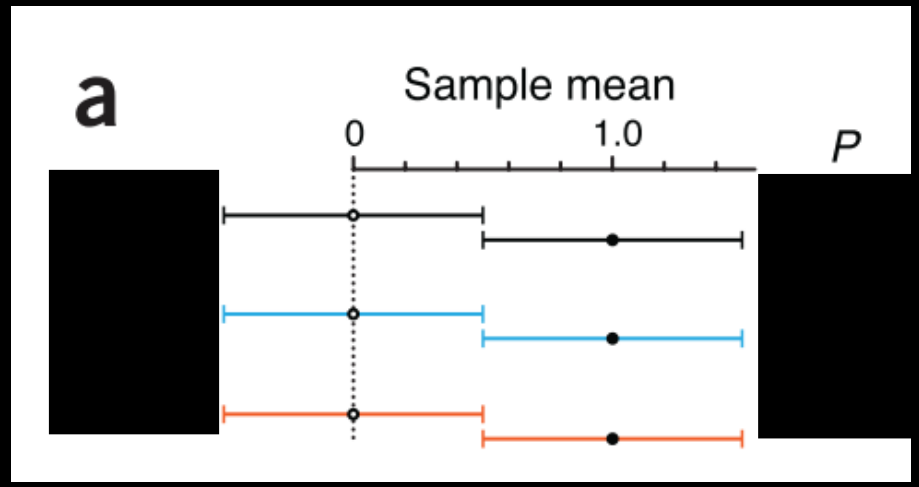


Great you have a mean from your measurements – Let's add some error bars

- Frequently, error bars are either:
  - Standard deviation (s.d.)
  - Standard error of the mean (s.e.m.)
  - X% Confidence Interval (X% CI)

- If you don't label your error bars, you're a heinous person.

Two samples of means 0 and 1.0, with 10 measurement each.



S.d. – The Good

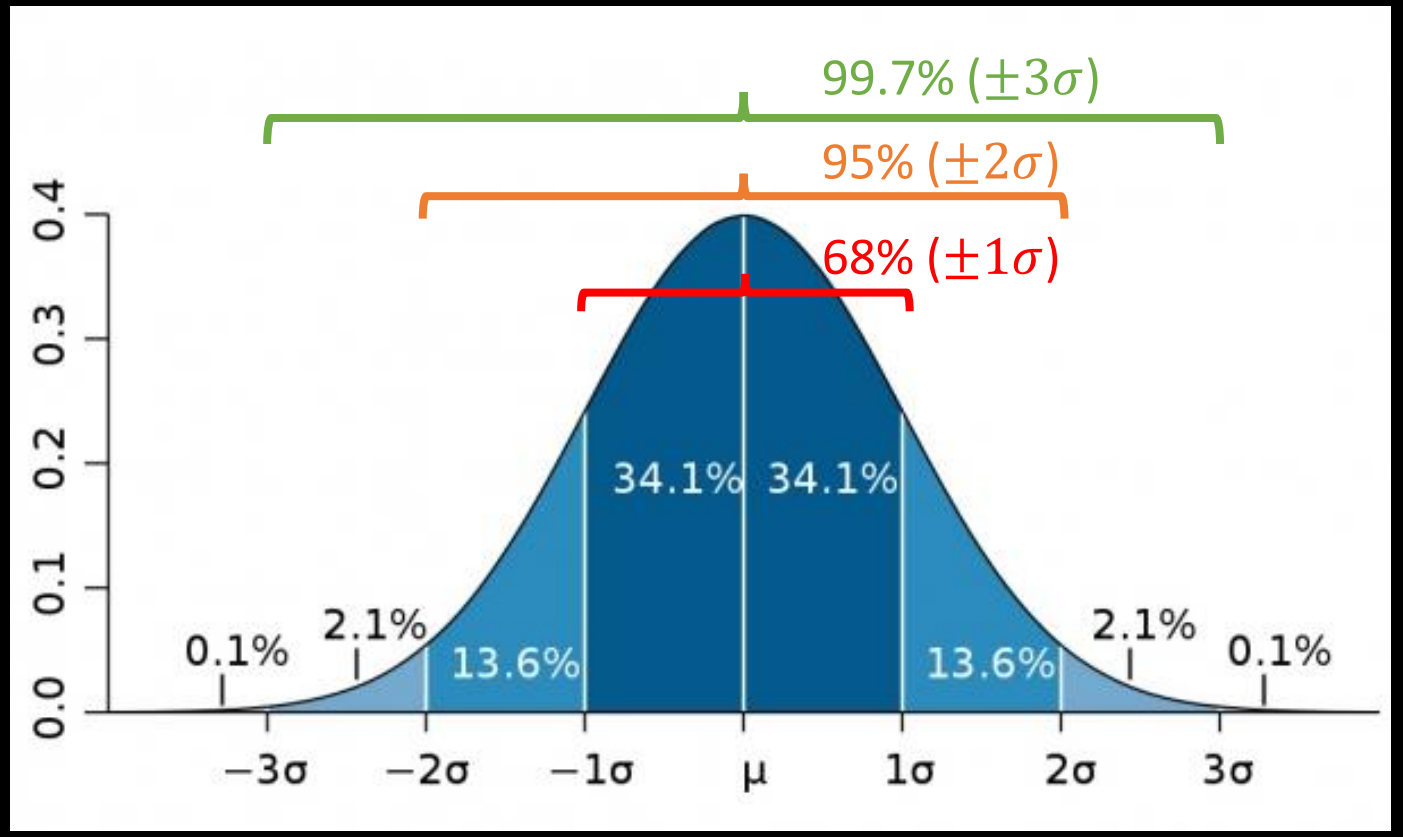
- The standard deviation shows the variation of the data.
- Helps identifying when there is overlap between two populations

S.d. – The Bad

We don’t intuitively understand what the s.d. represents quantitatively.

Only about 68% of the values (of a normally distributed population) fall within $\pm 1\sigma$

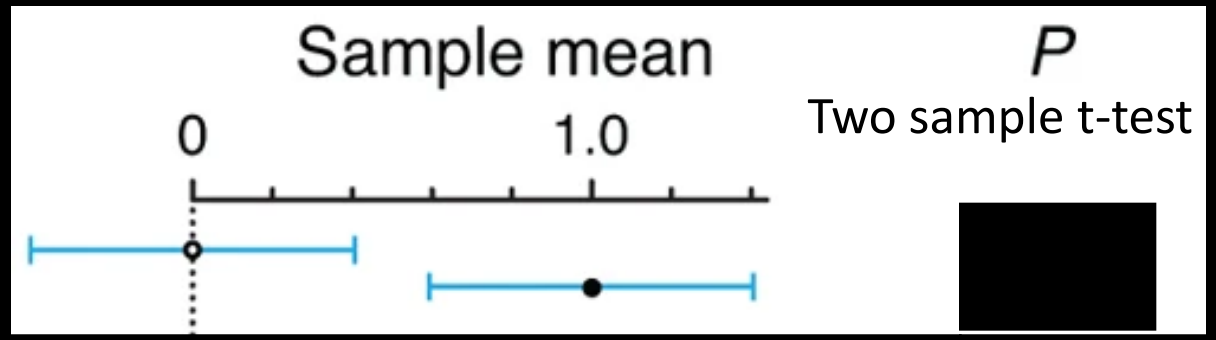
-> 32% of the values fall outside that range, but still belong to the same distribution.



S.e.m. bars – The Good

- Reflects the uncertainty on the mean.
- Intuitively decreases as the sample size increase.

« Hehe, I’m not an unsophisticated idiot, I don’t use the s.d. I use the s.e.m. »



Not even one star

*

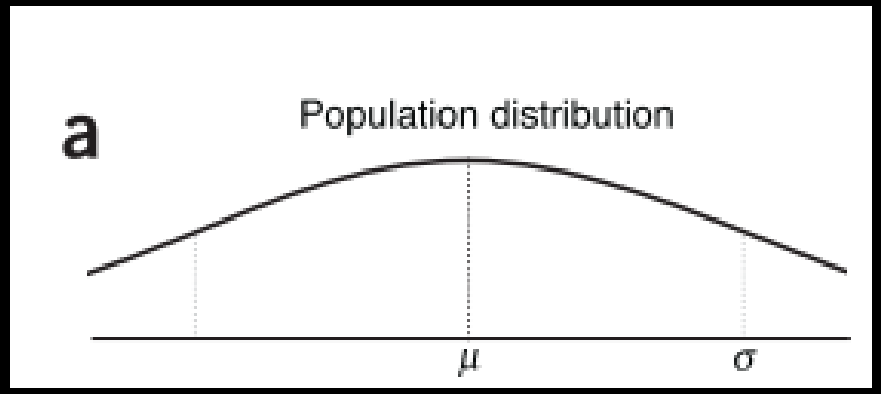
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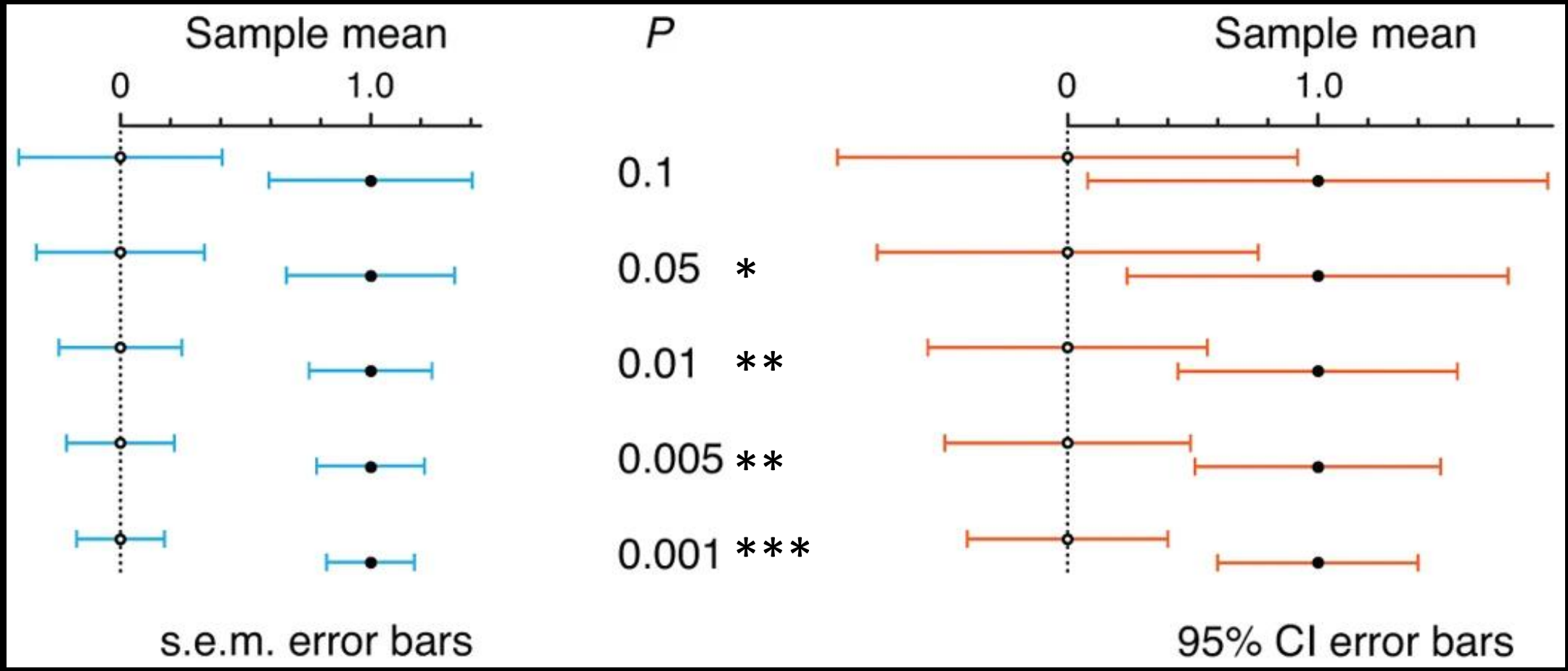
Confidence intervals (CI) – My beloved

- « If multiple samples were drawn from the same population, and a 95% CI calculated for each sample, we would expect the population mean to be found within 95% of the CIs. »



- Example: drawing N=20 samples (with n=10 points in each). On average, 95% of the intervals (19 out of 20) should capture the mean.
  - Here, *by chance*, 2 out of 20 missed the mean.

Confidence intervals



Comparing two populations: Why show confidence intervals and not just give a p-value?

- CI also « show » the statistical significance : when the CI bars just touch, this is « *highly* » statistically significant (**, P=0.005)

- More importantly, CI bars also show the magnitude of the variability -> your reader can then decide if the difference between two groups (e.g. control and treated) is RELEVANT.

Summary – Basics

STATISTICS GLOSSARY

Some common statistical concepts and their uses in analysing experimental results.

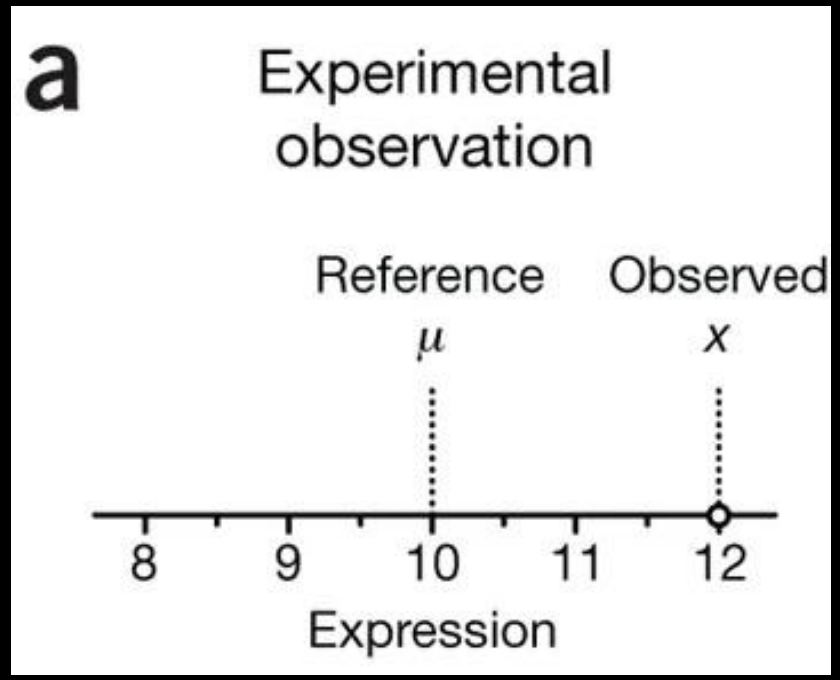
Term	Meaning	Common uses
Standard deviation (s.d.)	The typical difference between each value and the mean value.	Describing how broadly the sample values are distributed. $\text{s.d.} = \sqrt{(\sum (x - \text{mean})^2/(N-1))}$
Standard error of the mean (s.e.m.)	An estimate of how variable the means will be if the experiment is repeated multiple times.	Inferring where the population mean is likely to lie, or whether sets of samples are likely to come from the same population. $\text{s.e.m.} = \text{s.d.}/\sqrt{N}$
Confidence interval (CI; 95%)	With 95% confidence, the population mean will lie in this interval.	To infer where the population mean lies, and to compare two populations. $\text{CI} = \text{mean} \pm \text{s.e.m.} \times t_{(N-1)}$

From Research methods: Know when your numbers are significant, D. L. Vaux, *Nature* **492,** 2012

p values

And why you might want to reconsider

What is a *p* value – A western blot example



We measure an expression value of x = 12 for a *mutant case.*
From literature, past measurements, we know that the reference level of *wild type* is μ = 10.

Null hypothesis H_0 : let's suppose that our observation is part of the regular, *wild type* distribution of results.
We « know » this regular distribution through a large number of independent measurements.

The *p* value is the greyed area : the probability of sampling a value from the *wild type* distribution that is *at least as extreme as x.*

What is a *p* value

- « The *p* value is the probability, under the assumption that there is no true effect or no true difference, of collecting data that shows a difference equal to or more extreme than what you actually observed. »

- In the case of drug testing: « Even if my medication was *completely ineffective*, and only luck was at play, what are the chances my experiment would have produced the observed outcome ? »

- A *p* value is a measure of surprise. The smaller the *p* value, the more surprising the result is, in the context of the null hypothesis.

What a *p* value is NOT

- A measure of how right you are.
- A measure of the magnitude of the effect.
- A measure of how relevant the effect measured is.

- *p* values NEVER tell you that an hypothesis is correct or wrong. At best it *corroborates* an hypothesis.

Why focussing only on *p* values is Not A Good Idea® - Case study

- You are developing a drug to fight CAVID-23 and are looking for FDA approval.

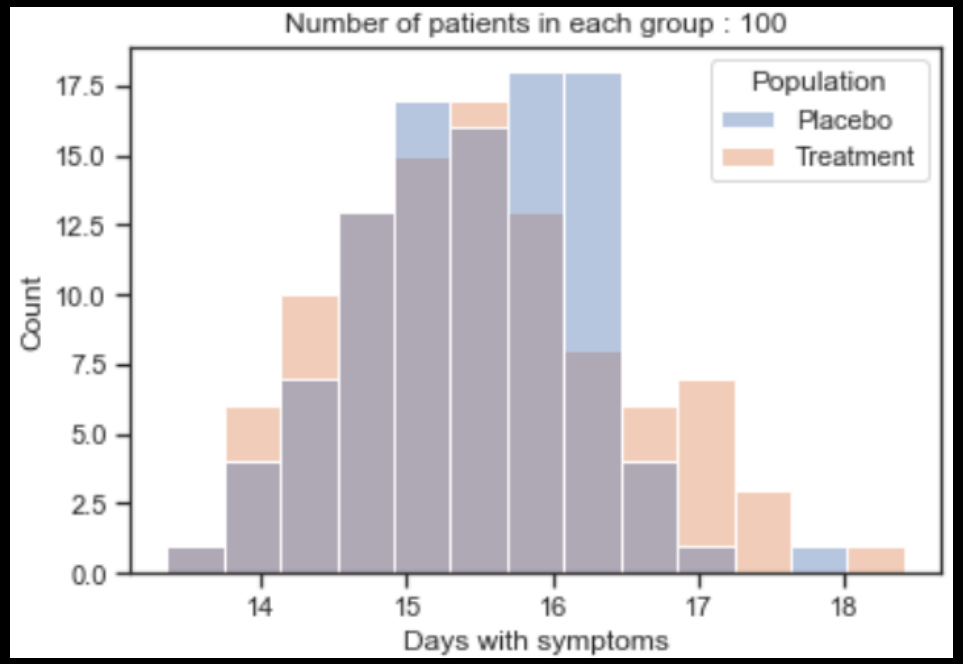
- Untreated - Average number of days with symptoms : 15.5

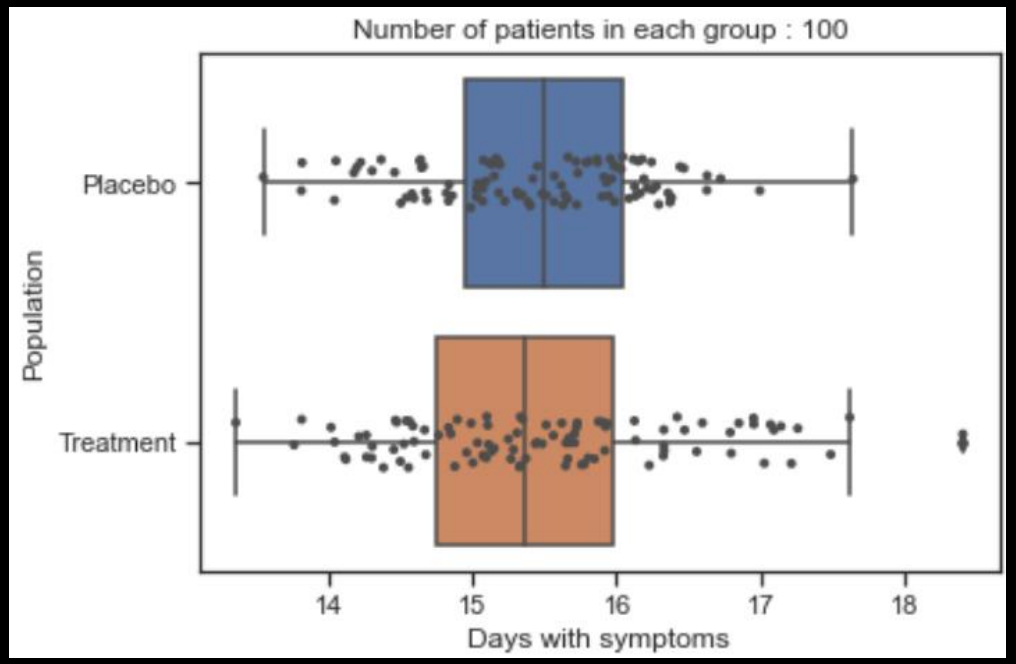
- The algorithm for approval is highly sophisticated

```
def fda_approval_decider(p_val, threshold=0.05):
    if p_val > threshold:
        print("Too bad buddy, your drug sucks, FDA approval denied")
    else:
        print("Let's gooooooooooooooooooooo, FDA approved!!!")
```

Example 0 - small effect, small sample size

- Your drug cuts the number of symptomatic days from 15.5 to 15.49 (!!!)
- Run a trial : 100 patients with placebo, 100 patients with drug.



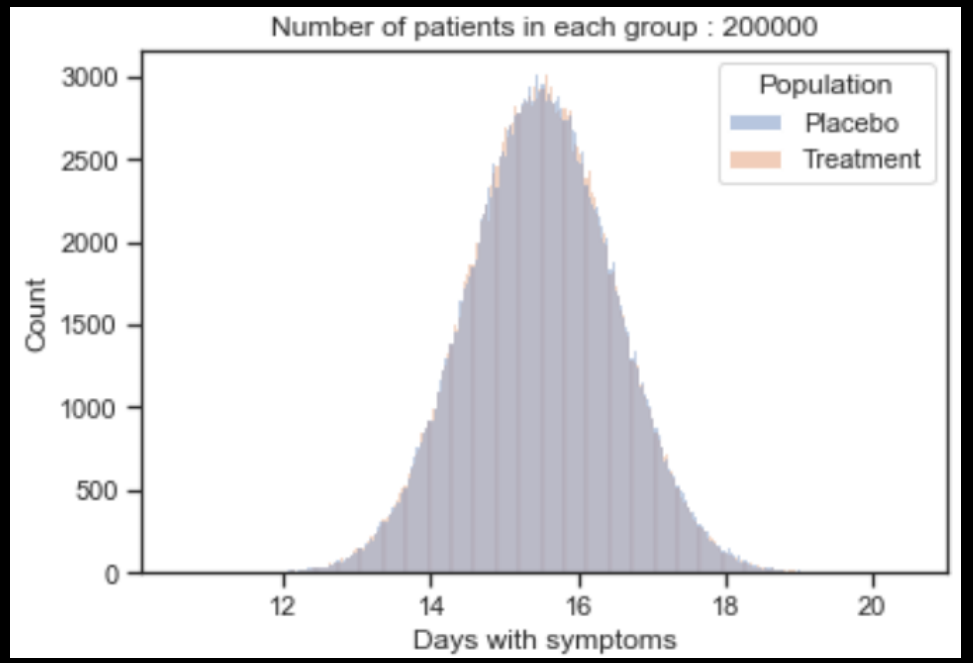


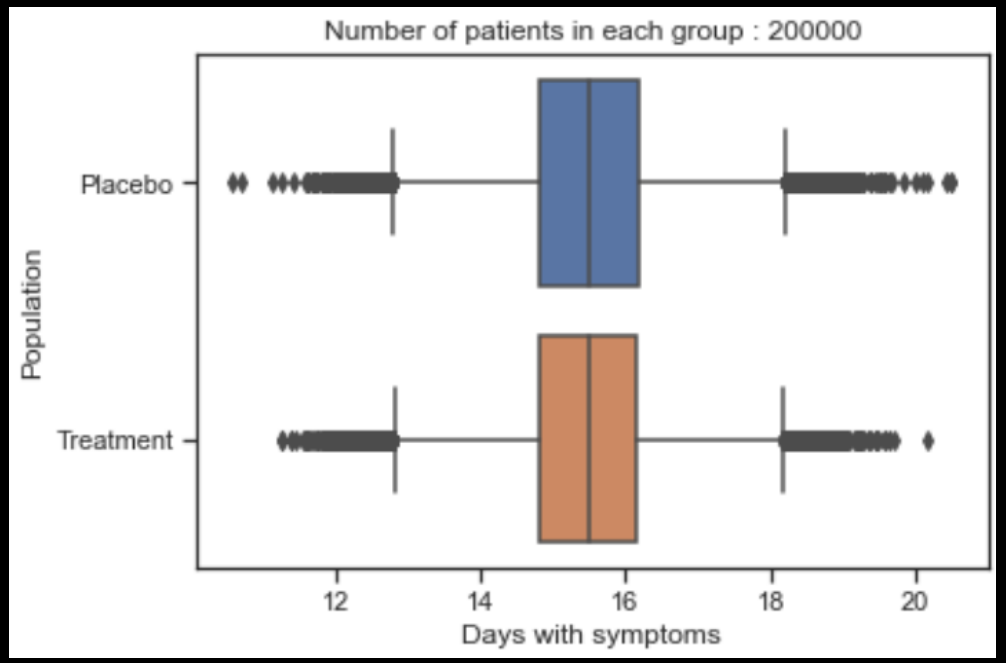
Median
1.5 IQR

```
The p value for a sample size of 100 in each group is 0.8128329083521515
Too bad buddy, your drug sucks, FDA approval denied
```

Example 1 - small effect, big sample size

- Your drug cuts the number of symptomatic days from 15.5 to 15.49 (!!!)
- Run a trial (but you're rich): 200 000 patients with placebo, 200 000 patients with drug.



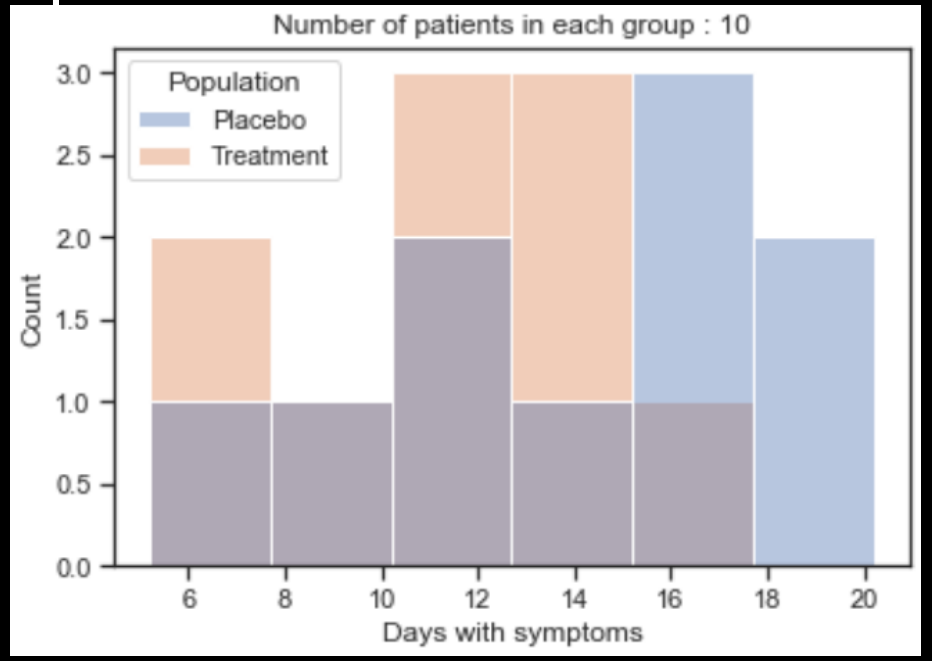


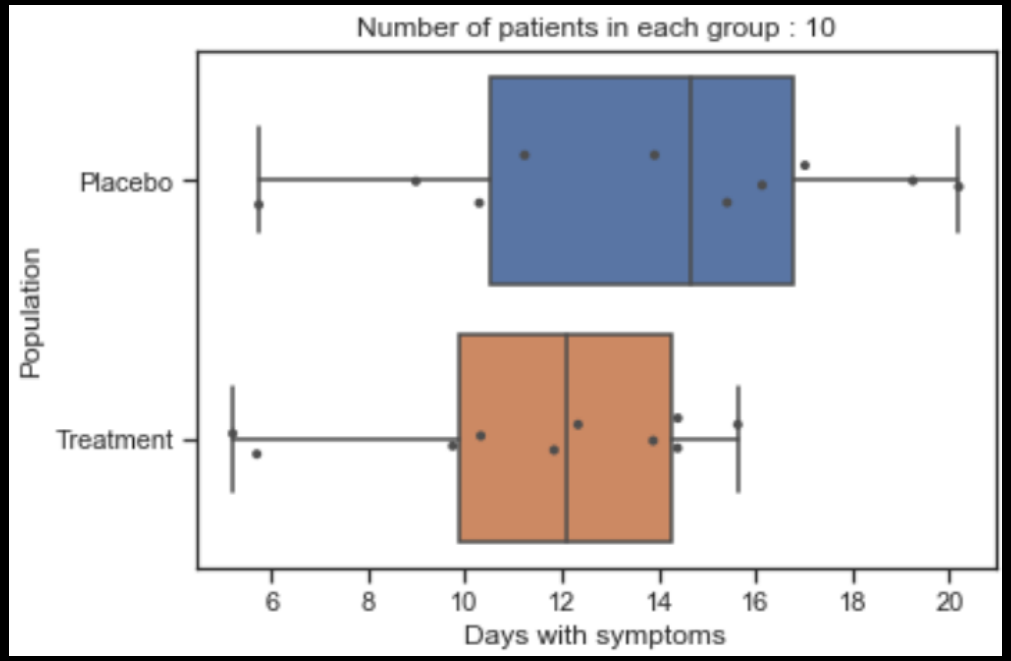
Median
1.5 IQR

```
The p value for a sample size of 200000 in each group is 0.0077473069182539 25
Let's gooooooooooooooooooooo, FDA approved!!!
```

Example 2 - big effect, small sample size

- Your drug cuts the number of symptomatic days from 15.5 to 10 (!!!)
- Run a trial (but you're poor) : 10 patients with placebo, 10 patients with





Median
1.5 IQR

```
The p value for a sample size of 10 in each group is 0.20189014528524135
Too bad buddy, your drug sucks, FDA approval denied
```

And if you think that, *surely*, no one would do this

Kairaluoma *et al.* – Pancreatic resection for carcinoma of the pancreas and the periampullary region in patients over 70 years of age, *Br J Surg* **74** 1987

Table 4 *Survival in relation to tumour histology*

Age	No. of patients	Mean ±s.d. (months)	Range (months)	Median (months)	Cumulative 5-year survival ±s.d. (%)
Adenocarcinoma of the pancreas					
⩾70 years	11	11±12	1–44	9	—
<70 years	24	11±10	1–44	8	—
Overall	35	11±11	1–44	9	—
Periampullary carcinoma					
⩾70 years	8	25±24	>10 years	20	25±22 } n.s.
<70 years	19	43±40	>10 years	36	58±19 }
Overall	27	34±32	>10 years	28	45±15

n.s., not significant

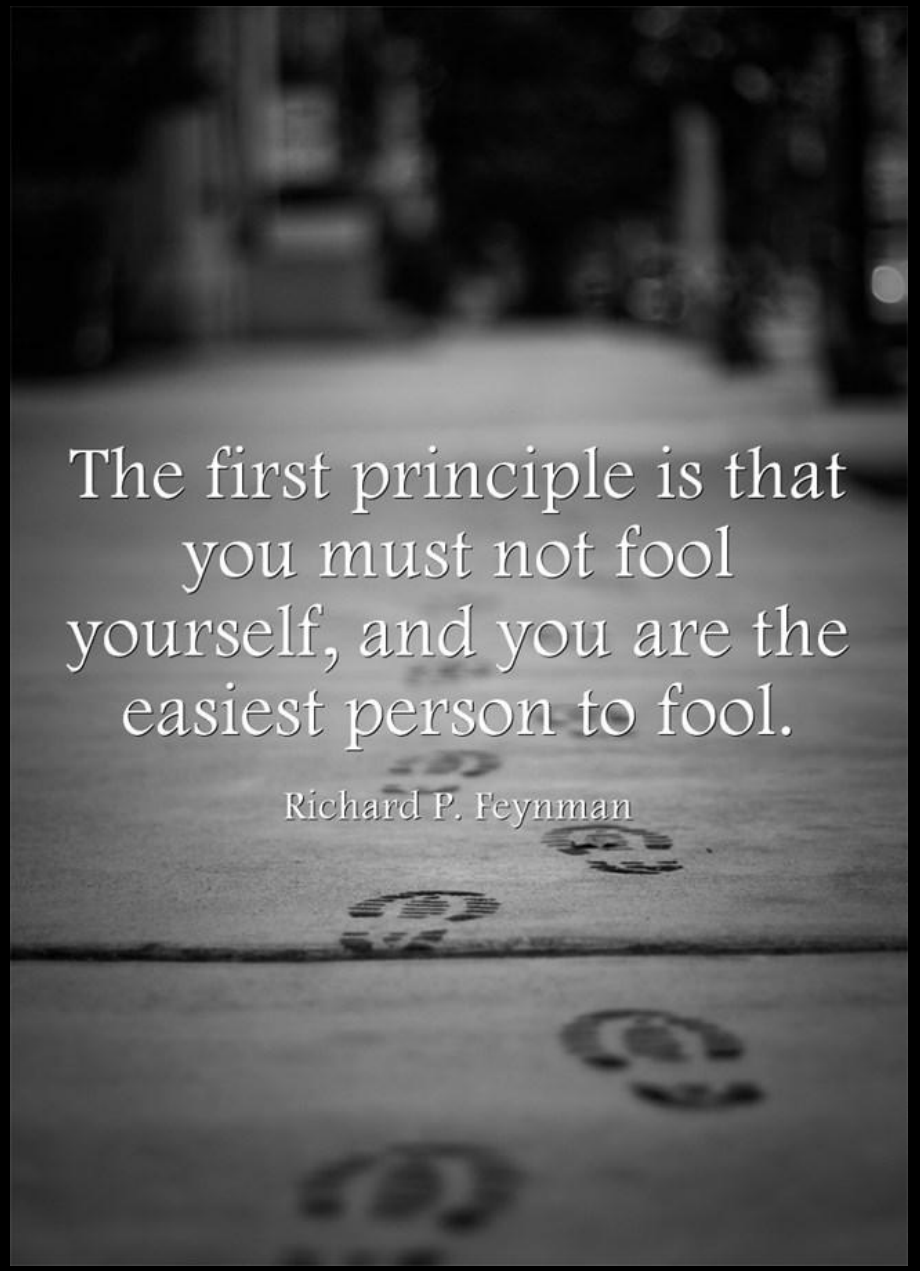
p = 0.05

« There were no significant differences between the groups. Of patients with periampullary carcinoma, those under the age of 70 fared better, but the difference was not significant.

[…]
In the present study the median and cumulative survival rates were similar in both groups, although patients with periampullary cancer who were under 70 fared somewhat better

[…] We conclude that age as such is not a limiting factor for pancreatic resection.»

In fairness to the authors : provided careful pre-operative selection, and low post-operative lethality, resection might still provide *clinically significant benefits* to the patients.



#hustle
#crush
#sigma_mentality

- A statistically *insignificant* result doesn’t mean there’s no difference between two groups.

- What about a significant result ?

Screening 100 drugs against cancer

TP	TP	TP	TP	TP	TP	TP	TP	FN	FN
FP	FP	FP	FP	FP	FP	TN			

TRUE POSITIVES **« There is an effect, and I correctly detect it »**	**TRUE NEGATIVES** **« There is no effect, and I correctly detect that there is no effect »**
FALSE POSITIVES **« There is no effect, BUT I detect that there is one »**	**FALSE NEGATIVES** **« There is an effect, BUT I missed it »**

- The **base rate of effective drugs** is 10%.
- Assuming a statistical power of 0.8 : of the 10 drugs that truly work (cyan + gold), I will only detect 8 of them (cyan).
- Assuming p = 0.05, I have a 5% chance of falsely concluding that an ineffective drug works. Out of the 90 that do NOT work (red + gray), I will incorrectly conclude that 5 of them are working (red)

- We conclude that there are 13 « working » drugs (8 cyan + 5 red)!
- The chance of a « working » drug being truly effective is 8/13 = 62%
- The *false discovery rate* is 5/13 = 38% !

False positive paradox - Careful with the base rate

- Low base rate, high false discovery rate -> many opportunities for false positives, even with a « significant *p* value ».

- Extreme case : batch of 100 drugs, with a base rate of 0%, no shot at getting a truly working drug.
  - Nevertheless you can still get $p < 0.05$ for 5% of the drug in the batch.

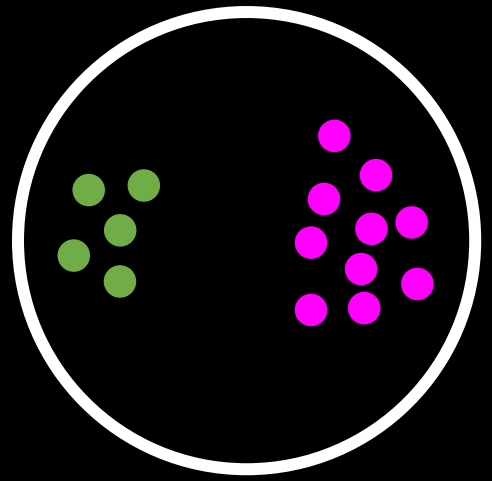
For illustrative purposes only, don't @ me

The base rate fallacy applied to COVID

Hospitalized with COVID at Hospital A

Un-vaccinated

Vaccinated



5 un-vaccinated versus 10 vaccinated

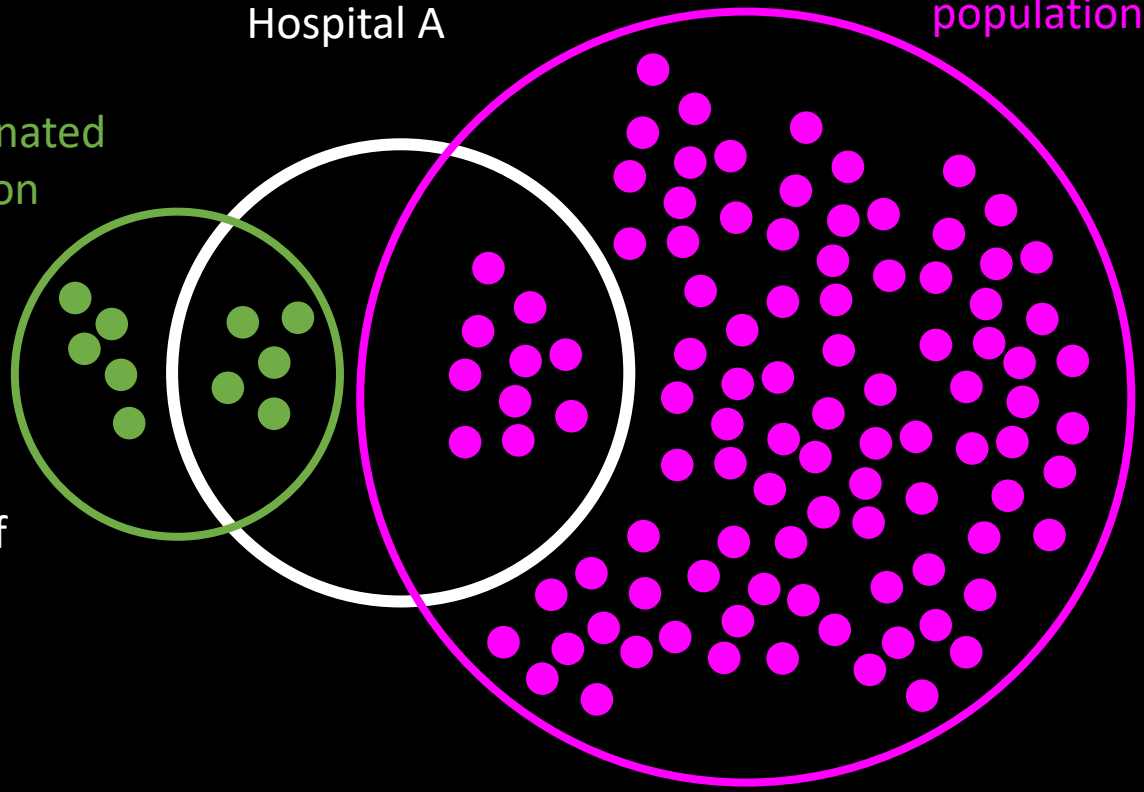
Worrisome, the vaccine is a hoax, COVID is a hoax, etc., etc.

However, this ignores the base rate of each group in the TOTAL population

Hospitalized with COVID at Hospital A

Un-vaccinated population

Vaccinated population



Rate of hospitalized & un-vaccinated : 5/10 = 50%

Rate of hospitalized & vaccinated : 10/100 = 10%

Adapted from https://twitter.com/MarcRummy/status/1464178903224889345

Summary

- Don’t just report the p-values
- Statistical significance is not clinical relevance.
- Absence of statistical significance is not absence of effect
- Be mindful of the base rates

(Under)Power

Sample size effects, and a small trick to help you

(Under)Power

- Suppose your drug cuts the risk of heart attack from 20% to 10%.
- To evidence a 0.1 effect size, with 80% certainty (the power of the test), at p=0.05, what sample size would you need ?

https://tweedback.de/z4zj

G.D. Murray – The task of a statistical referee, *British Journal of Surgery* **75**, 1988

(Under)Power

- Suppose your drug cuts the risk of heart attack from 20% to 10%.
- To evidence a 0.1 effect size, with 80% certainty (the power of the test), at p=0.05, what sample size would you need ?

```
from statsmodels.stats.power import TTestPower
```

```
power = TTestPower()
n_test = power.solve_power(nobs=None, effect_size=0.1, power = 0.8, alpha=0.05)
```

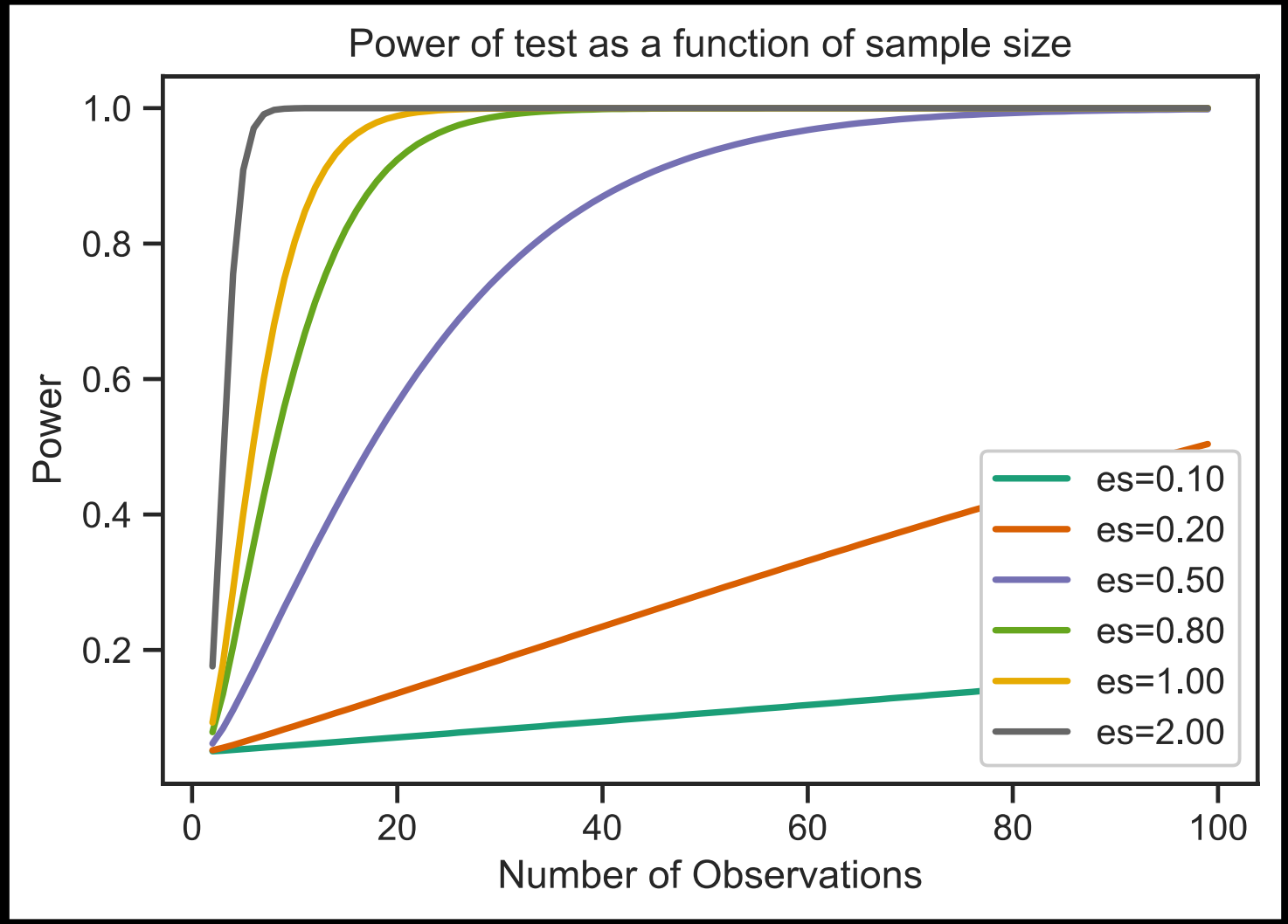
```
To identify a 0.1 effect size, with 80% certainty, and p=0.05, you need a sample size of: 393.404 - One-sided
```

G.D. Murray – The task of a statistical referee, *British Journal of Surgery* **75**, 1988

Beware the underpowered study

- Small effect sizes require a large number of observations.
- Huge effect sizes require less

es : effect size



We rarely have the luxury of 10 000 samples

Suppose that our « true population » is normally distributed with $\mu = 5.0$ and $\sigma = 2.0$ but we don't know it.

Let's run an experiment with n = 10

```
[5.60943416 2.92003179 6.50090239 6.88112943 1.09792962 2.39564099
 5.25568081 4.36751482 4.96639768 3.29391214]
Initial sample mean is : 4.329
```

```
[6.75879595 6.55558387 5.1320614  7.25448241 5.93501868 3.28141507
 5.73750157 3.0822348  6.7569006  4.90014818]
If we drew again, the mean of samples_2 could be : 5.539
```



~~There's 24 hours in a day, just repeat it !~~

Bootstrapping ! – Simulating more samples

Our « true population » is normally distributed with $\mu = 5.0$ and $\sigma = 2.0$

```
[5.60943416 2.92003179 6.50090239 6.88112943 1.09792962 2.39564099
 5.25568081 4.36751482 4.96639768 3.29391214]
Initial sample mean is : 4.329
```

```
[2.39564099 5.60943416 4.96639768 2.92003179 2.39564099 6.50090239
 6.88112943 2.39564099 4.96639768 5.25568081]
Bootstrap sample 1 mean is : 4.429
```

```
[2.39564099 1.09792962 6.88112943 3.29391214 6.88112943 5.60943416
 3.29391214 3.29391214 2.39564099 2.92003179]
Bootstrap sample 2 mean is : 3.806
```

Bootstrapping !

Our « true population » is normally distributed with $\mu = 5.0$ and $\sigma = 2.0$

```
## Repeat this operation - a lot

bs_sample_means = []
number_of_replicates = 1000

for i in range(number_of_replicates):
    bs_sample = np.random.choice(samples, replace=True, size=10)
    bs_sample_means.append(np.mean(bs_sample))
```

```
# The 95% confidence interval is now located between the 2.5 and 97.5 percentiles
```

```
The 95% confidence interval for the mean : [3.225 ; 5.438]
```

Summary

- Our intuition about what makes « a reasonable number of patients » is probably wrong.
- To evidence small effect sizes you need *a lot* of observations.
- Bootstrapping can help you estimate a confidence interval if you cannot generate an infinite number of samples.

Solutions to the quizz

From Haller & Krauss

The problem

- Suppose you're testing two drugs, Fixitol and Solvix.
- You have two treatment groups, one that takes Fixitol and one that takes Solvix, and you measure their performance on a standard fitness test afterward.
- You compare the mean score of each group using a simple significance test, and you obtain $p = 0.01$, indicating there is a statistically significant difference between means.

Answer

- Everything is « False »

- Q1 - You have absolutely disproved the null hypothesis ("There is no difference between means")
- Q3 - You have absolutely proved the alternative hypothesis ("There IS a difference between means")

- Significance tests ***cannot*** prove/disprove hypotheses. They provide probabilistic information that can corroborate theories.

- Q2 - There is a 1% probability that the null hypothesis is true
- Q4 - You can deduce the probability that the alternative hypothesis is true.

- Generally, you cannot(*) assign a probability to a hypothesis using a significance test (neither ‘‘1’’, nor anything).

- (*) you could, using Bayesian statistics.

- Q5 - You know, if you decide to reject the null hypothesis, the probability that you are making the wrong decision.

- A bit subtle. This looks like a « False Negative » definition. But having actually rejected the H_0, this decision would be wrong, if and only if the H_0 were true.
- Thus, the probability in Q5 ("...that you are making the wrong decision") is $p(H_0)$ and this probability – as we learned from Q2 – cannot be derived with a null hypothesis significance testing.

- Q6 - You have a reliable experimental finding, in the sense that if your experiment were repeated many times, you would obtain a significant result in 99% of trials.

- Statement 6 reflects the so-called “replication fallacy”. In Neyman and Pearson’s paradigm, one could interpret α = .01 in a frequentistic framework as relative frequency of rejections of H_0 if H_0 is true. The example however gives no evidence of H_0 being true.

Final prescriptions

If the results from one representative experiment are shown, then N = 1 and statistics do not apply.



D. L. Vaux

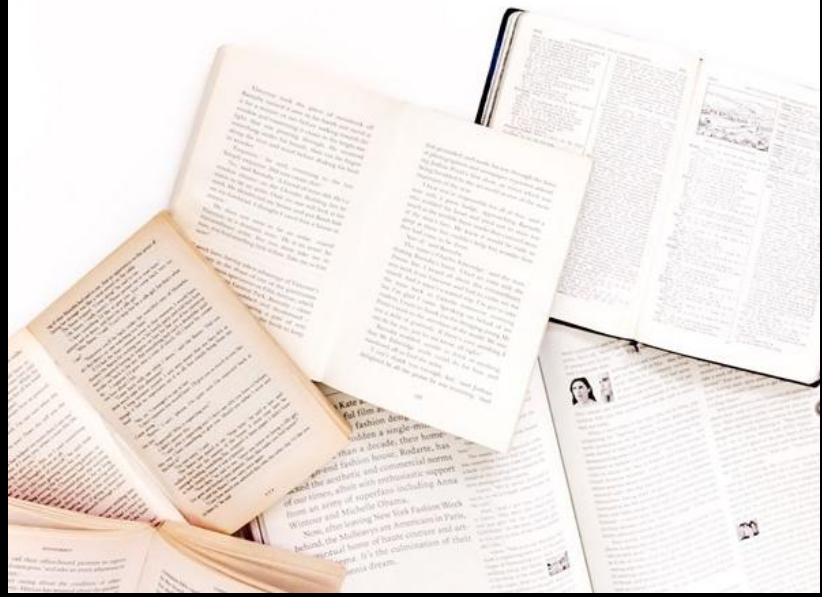
When N is only 2 or 3, it would be more transparent to just plot the independent data points, and let the readers interpret the data for themselves.

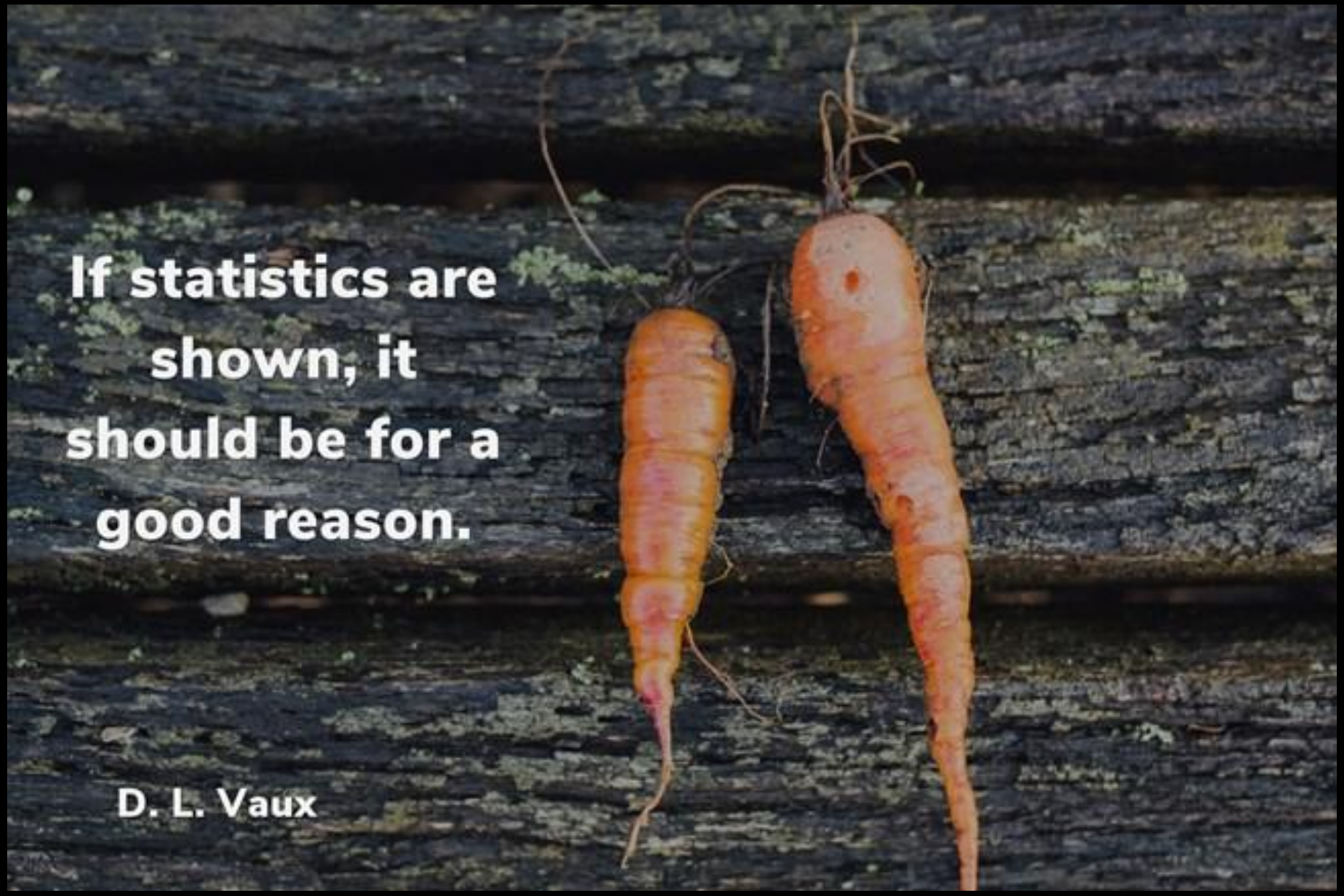
D. L. Vaux



If the data in an experiment are equivocal, or the effect size is small, it is much better to come up with an extra, mechanistically different, experiment to test the hypothesis, than to repeat the same experiment until P is less than 0.05.

D. L. Vaux




If statistics are shown, it should be for a good reason.
D. L. Vaux

Don't be blinded by stats.
83% of the people survive a round of russian roulette.
That does not make it a safe game.



S. R.

Journal Guidelines

Nature

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☐	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☐	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☐	A description of all covariates tested
☐	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☐	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☐	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

PLOS ONE

Reporting of statistical methods

In the methods, include a section on statistical analysis that reports a detailed description of the statistical methods. In this section:

- List the name and version of any software package used, alongside any relevant references
- Describe technical details or procedures required to reproduce the analysis
- Provide the repository identifier for any code used in the analysis (See our code-sharing policy.)

Statistical reporting guidelines:

- Identify research design and independent variables as being between- or within-subjects
- For pre-processed data:
  - Describe any analysis carried out to confirm the data meets the assumptions of the analysis performed (e.g. linearity, co-linearity, normality of the distribution).
  - If data were transformed include this information, with a reason for doing so and a description of the transformation performed
- Provide details of how outliers were treated and your analysis, both with the full dataset and with the outliers removed
- If relevant, describe how missing/excluded data were handled
- Define the threshold for significance (alpha)
- If appropriate, provide sample sizes, along with a description of how they were determined. If a sample size calculation was performed, specify the inputs for power, effect size and alpha. Where relevant, report the number of independent replications for each experiment.
- For analyses of variance (ANOVAs), detail any post hoc tests that were performed
- Include details of any corrections applied to account for multiple comparisons. If corrections were not applied, include a justification for not doing so
- Describe all options for statistical procedures. For example, if t-tests were performed, state whether these were one- or two-tailed. Include details of the type of t-test conducted (e.g. one sample, within-/between-subjects).
- For step-wise multiple regression analyses:
  - Report the alpha level used
  - Discuss whether the variables were assessed for collinearity and interaction
  - Describe the variable selection process by which the final model was developed (e.g., forward-stepwise; best subset). See SAMPL guidelines.
- For Bayesian analysis explain the choice of prior trial probabilities and how they were selected. Markov chain Monte Carlo settings should be reported.

References

- Statistics Done Wrong: The Woefully Complete Guide – Alex Reinhart, No Starch Press 2015
- Research methods: Know when your numbers are significant – David L. Vaux, Nature 2012, 10.1038/492180a
- The whole « Point of significance: Statistics for Biologists » collection: https://www.nature.com/collections/qghhqm/pointsofsignificance
- The task of a statistical referee – G.D. Murray, *Br J Surg* **75** 1988

Further reading

- Ten common statistical mistakes to watch out for when writing or reviewing a manuscript – Makin & Orban de Xivry, eLife **8** 2019
  - Highlights problems, how to detect them, and solutions for researchers
- Datacamp – Statistical Thinking in Python (Part 1 & 2) and Case Studies in Statistical Thinking
  - For a very hands-on approach around case studies

Additional notes

- Get a pro BEFORE having to do stats.
- Don’t do stats when you can. Get a pro.
- Don’t PRETEND to be able to do stats. Get a pro.
- An external statistician needs background knowledge about your issue. A long term relationship is better for everyone.
- « Describe the results in terms of a number which has a clear clinical interpretation rather than a P value which in isolation is meaningless. »

Additional points

- « The mean and, in particular, the standard deviation, are only useful summaries when the data are symmetrically distributed. »
- « Many variables encountered in medicine are skewed, with all of the ‘abnormal’ values lying to one side of the distribution. »
- A Student t-test does not make sense on a non-normal/very skewed distribution.